

An Approach for Applying Cultural Study Theory to Technical Writing Research

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When the idea of culture is expanded to include institutional relationships extending beyond the walls of one organization, technical writing researchers can address relationships between our power/knowledge system and multiculturalism, postmodernism, gender, conflict, and ethics within professional communication. This article contrasts ideas of culture in social constructionist and cultural study research designs, addressing how each type of design impacts issues that can be analyzed in research studies. Implications for objectivity and validity in speculative cultural study research are also explored. Finally, since articulation of a coherent theoretical foundation is crucial to limiting a cultural study, this article suggests how technical writing can be constituted as an object of study according to five (of many possible) poststructural concepts: the object of inquiry as discursive, the object as practice within a cultural context, the object as practice within a historical context, the object as ordered by language, and the object in relationship with the one who studies it.

"Merely qualitative knowledge leads to the grossest errors of judgment, and is of that kind of little learning which is a dangerous thing. . . . All engineering problems are purely quantitative from the beginning to the end, and so are all other problems, whether material, or moral, or financial, or commercial, or social, or political, or religious. All judgments passed on such problems, therefore, must be quantitative judgments. How poorly prepared to pass such judgments are those whose knowledge is qualitative only!"
John Butler Johnson, 1928

"Once every text had to take account of the theologians and the authorities, so today every text is written against the back-drop of science and the authority of truth with a small t. . . . And without this truth the distinctions between science and art, between literature and philosophy, between stories and truths, begin to look precarious."
Hilary Lawson, 1989

Good technical writing is so clear that it is invisible. Yet technical writing is the mechanism that controls scientific systems, thereby organizing the operations of modern institutions and the people within them. The invisibility of technical writing attests to its efficiency as a control mechanism because it works to shape our actions without displaying its methods for ready analysis. When technical writing is made visible for study, it is often characterized as a simple collaborative effort in which writers mediate technology for users. Yet it can also be seen as a mundane discourse practice working to legitimate some types of knowledge while marginalizing other possible knowledges. These questions of knowledge legitimation within cultural contexts extending beyond the walls of one organization cannot be addressed well through social constructionist research designs. They can be addressed, however, through cultural study.

During the course of the twentieth century, our concepts of technical writing have developed in tandem with other elements of our culture. Like any other aspect of our culture, therefore, technical writing can be fruitfully explored through a cultural studies research framework. This article will explore how cultural studies are helping to illuminate technical writing issues left unseen in more scientifically modeled research studies—issues such as effects of institutional relationships and expanded notions of culture on technical writing practices, as well as the cultural power of knowledge produced through technical writing. This article will also discuss some implications of using a speculative research model in institutions where non-scientific knowledge is often seen as non-legitimate. Finally, this article will offer some guidelines for designing a cultural study of technical writing, focusing on how to limit the study and providing some theoretical reasoning for this delimiting process.

Putting Technical Writing Practices in Cultural Contexts

Researchers in technical writing have begun to explore how technical writing is involved within situated institutional relationships of knowledge and power—how some types of knowledge are legitimated through technical writing practice, while other possible knowledges (such as knowledge gained through speculation, emotion, or intuition) are subjugated or excluded as marginal. In order to carry out such a critical analysis of technical writing, institutions where technical writing is practiced need to be reconstructed as cultural agents which are not necessarily bounded by any one organization's walls. Vincent Leitch described how institutions as cultural agents influence discourse:

Through various discursive and technical means, institutions constitute and disseminate systems of rules, conventions, and practices that condition the creation, circulation, and use of resources, information, knowledge, and belief. Institutions include, therefore, both material forms and mechanisms of production, distribution and consumption and ideological norms and protocols shaping the reception, comprehension, and application of discourse.

. . . Institutions often enable things to function, inaugurate new modes of knowledge, initiate productive associations, offer assistance and support, provide useful information, create helpful social ties, simplify large-scale problems, protect the vulnerable, and enrich the community. (127-28)

Because institutions are cultural agents that affect discourse practices, a study of technical writing as situated in systems of knowledge and power is incomplete if the idea of culture is limited within one organization. Recognition of an organization's participation in cultural contexts enables a critical study that can illuminate assumptions about the inevitable roles of technical writing in our culture at given historical moments—roles such as information conveyor or technology mediator.

A view of culture that is limited within the walls of one organization does not allow researchers to question assumptions about technical writing practices because those practices are not placed in relationship to influences outside the organization under study. This limited view of culture is tacitly adopted in traditional social constructionist research designs, however, as illustrated by Stephen Doheny-Farina's exploration of technology transfer, *Rhetoric, Innovation, Technology*. In this study, technical writing is seen as participating in a "series of personal constructions and reconstructions of knowledge, expertise, and technologies by the participants attempting to adapt technological innovations for social uses" (ix). Although Doheny-Farina places technical writing practice in a social setting, he assumes that problems in technology transfer have more to do with the quality of the technical texts produced by one organization than with economic, political, or social pressures or conflicts affecting a situated writing practice. Doheny-Farina could describe how a text shapes practice, but does not question why the text includes the information that it does and not other information that would be equally possible to include—why the text legitimates some kinds of knowledge and not others. And further, what systems of power (the cultural context of the organization under study) does the knowledge legitimated in the text uphold and what other possible systems of power did the text make impossible? These research questions could not be examined using a social constructionist research design that limited the idea of culture within one organization. To answer questions about knowledge legitimation would require an assumption that the culture within the organization under study was in tension with a wider culture that included related institutions.

Similarly, when Doheny-Farina analyzes James Paradis, David Dobrin, and Richard Miller's study of "Writing at Exxon ITD," he finds that their writing practice "becomes part of the process of developing an organizational identity; it becomes part of the process of group membership" (*Rhetoric* 28). Social influences which form the culture under investigation in this analysis of the Exxon study are limited to the interior of the Exxon Corporation. The "social" in "social constructionist" here extends to a group of people within one organization. Because of this limited view of culture, Doheny-Farina is not able to explore how Exxon's texts interact with texts from competing companies, government agencies, and the legal system, for example.

The limitations of confining culture within the walls of one organization are further illustrated in Doheny-Farina's consideration of a 1991 article by Carl Herndl, Barbara Fennell, and Carolyn Miller in which these authors examine miscommunication and misunderstanding in the Three Mile Island and *Challenger* disasters. Doheny-Farina constructs the communication problems that contributed to the *Challenger* disaster as problems of technology mediation: "Issues of texts miscommunication and misunderstanding . . . involve the failures of texts to mediate technology to users" (*Rhetoric* 28). This focus on the technical writer's role as mediator between technicians and users allows Doheny-Farina to explore how technical writers uphold science and technology's dominant position within our culture. But it does not question what influences other than miscommunication might have contributed to the misunderstandings described in the Herndl, Fennell, and Miller article. For example, the decision to go ahead with the *Challenger* launch was made despite evidence of previous O-ring erosion and a forecast of record cold weather at launch. Morton Thiokol engineer Roger Boisjoly argued for safety and against the launch at a meeting with NASA representatives the night before the *Challenger* was launched. He later suggested an extra-textual factor creating pressure to launch: Morton Thiokol was in the process of negotiating a \$1 billion contract with the U.S. government for space shuttle parts and the government was considering second-sourcing Morton Thiokol. NASA staff were intent on launching and, during the pre-launch meeting, asked Morton Thiokol management to "rethink" their recommendation not to launch. Morton Thiokol reversed its no-launch recommendation to acquiesce to NASA's wishes, despite Morton Thiokol engineers' warnings about unsafe launch conditions. More than a simple case of misunderstanding, discourse surrounding the decision to launch the *Challenger* was influenced by economic and political considerations that legitimated some knowledge (the managers' judgment and decision to launch) and marginalized other knowledge (the engineers' data and warnings). The knowledge legitimated in this case was clearly participating in systems of institutional power which extended far beyond Morton Thiokol's walls.

A limited view of culture is not unusual in technical and professional communication research. (See the following examples of social constructionist research which limit the idea of culture to one organization: Jennie Dautermann, "Negotiating Meaning in a Hospital Discourse Community;" Amy Devitt, "Intertextuality in Tax Accounting;" James Porter, "The Role of Law, Policy, and Ethics in Corporate Composing;" Graham Smart, "Genre as Community Invention;" Gail Stygall, "Texts in Oral Context;" and Charlotte Thralls, "Rites and Ceremonials.") In an example of separating a government agency's discourse practices from their context of political influences, Susan Kleimann assumes that culture resides within one government organization (the General Accounting Office) in her study titled "The Reciprocal Relationship of Workplace Culture and Review." Kleimann confines the idea of culture either to divisions within the General Accounting Office (GAO) or to the GAO as an isolated governmental entity and describes how the organization's values influenced document review practices:

Two major influences shape aspects of GAO's culture. First, GAO reports often result in changes to national policy, legislation, and funding; second, until recently, most GAO employees were educated as accountants, valuing minutiae and accuracy. Consequently, the agency has a cautious culture that demands maintaining detailed and extensive workpapers, referencing all facts to these workpapers, wanting both accuracy and objectivity and requiring an extensive review process. Moreover, because of the reports' potential impact, review is used to ensure that reports are correct from several perspectives. (58)

While Kleimann notes that GAO texts are influential in shaping national policy, legislation, and funding decisions, she confines the culture discussed in her study within the boundaries of the GAO. While she studies one of the major institutions in any culture (the government), she does not ask how national and international political tensions shape GAO texts and how politics are in turn shaped by GAO texts.

The limited and conservative view of culture found in much research in technical and professional communication is articulated in Jack Selzer's description of intertextuality: "Indeed, 'context' or 'environment' or 'setting' or 'culture' might be understood as nothing more than a complex of language and texts, and individuals within an environment therefore might be understood as minds assimilated into its concepts and terminology" (172). Or further:

Readers and writers experience the flow of culture as a kind of collaboration among seen and unseen authors and texts and readers, all in the process of making sense. One job of the critic of culture—of the rhetorician, that is—is to uncover the various resonances inscribed in the tapestry of text and to account for their source, their intricacy, and their meaning. (179)

While Selzer's view of culture acknowledges that cultural criticism involves itself with meaning, his view does not involve texts in tensions within situated relations of power and knowledge—where knowledge legitimation is contested by various interest groups and “making sense” means something different depending on your point of view. To view culture as a “kind of collaboration” works to sanitize what Walter Benjamin describes as barbarism inherent in the spoils of war (for cultural legitimation):

Whoever has emerged victorious participates to this day in the triumphal procession in which the present rulers step over those who are lying prostrate. According to traditional practice, the spoils are carried along in the procession. They are called cultural treasures. . . . There is no document of civilization which is not at the same time a document of barbarism. And just as such a document is not free of barbarism, barbarism taints also the manner in which it was transmitted from one owner to another. (“Theses” 256)

“Making sense” within a framework of contests for knowledge legitimation is not merely a “kind of collaboration.” From a critical point of view, making sense for the victor is not making sense for the vanquished, who might ask why their knowledge was silenced.

In *The Differend*, Jean-François Lyotard describes the silencing of non-legitimated knowledge as a “wrong” suffered in “a case of conflict, between (at least) two parties, that cannot be equitably resolved for lack of a rule of judgment applicable to both arguments” (xi). Because there is no universal rule for equitable judgment, actions taken through discourse must privilege one way of knowing over other possible ways of knowing. Unlike a simple idea of consensus-based collaboration, Lyotard's theory of discourse production holds that power is distributed unevenly among possible ways of knowing. Basing his description on how phrases are linked in discourse, Lyotard finds, “In the absence of a phrase regimen or of a genre of discourse that enjoys a universal authority to decide, does not the linkage (whichever one it is) necessarily wrong the regimens or genres whose possible phrases remain unactualized?” (xii). Using Lyotard's theory, discourse becomes a contest for legitimating knowledge and culture is more hegemonic than simply collaborative. Discourse becomes a struggle mediated by culture.

Struggles in discourse are contained within the discourse. For Michel Foucault, discourse holds histories of struggles for knowledge legitimation and the articulated discourse subsumes other discourses that were possible, but not articulated. In arguing for the study of culture through discourse analysis, Foucault describes how the legitimated knowledges articulated in discourse embody historical struggles for their legitimation and conquest:

In the two cases—in the case of the erudite as in that of the disqualified knowledges—with what in fact were these buried, subjugated knowledges really concerned? They were concerned with a *historical*

knowledge of struggles. In the specialized areas of erudition as in the disqualified, popular knowledge there lay the memory of hostile encounters which even up to this day have been confined to the margins of knowledge. (*Power/Knowledge* 83, emphasis in original)

At the margins of knowledge we can find two types of delegitimated knowledges: erudite learning that may have been previously legitimated knowledge, but has been subsumed (or conquered) by other subsequently legitimated knowledges; and naive "know-how" (de Certeau, *Practice* 65 ff.) that was previously legitimated as sufficient for carrying out everyday practices, but also has been subsumed by other subsequently legitimated knowledges (usually some sort of science or theory).

Because technical writing participates in institutional relationships, it works to organize knowledge through science and practice through theory. For example, because technical writing knowledge is made through institutions such as schools and publishing companies, naive or uneducated technical writing practices are organized through academic and economic systems which tend to reproduce our culture's dominant scientific model. This organizing activity is found by Michel de Certeau to be a trend in Western culture since the time of Francis Bacon: "the sciences are the operational languages whose grammar and syntax form constructed, regulated, and thus writeable, systems; the arts are techniques that await an enlightened knowledge they currently lack" (*Practice* 66). Because science forms the legitimated language of practice, it is a "writing that conquers" (*Writing of History* xxv) other practices based on naive "know-how:"

But at the same time that they acknowledge in these practices a kind of knowledge preceding that of the scientists, they have to release it from its "improper" language and invert into a "proper" discourse the erroneous expression of "marvels" that are already present in everyday ways of operating. Science will make princesses out of all these Cinderellas. The principle of an ethnological operation on practices is thus formulated: their *social isolation* calls for a sort of "education" which, through a *linguistic inversion*, introduces them into the field of scientific written language. (*Practice* 67, emphasis in original)

If technical writing is the mediator between technology and what we have come to term "users," technical writing practices work to conquer users' naive know-how and reformulate these uneducated practices into scientific discourse that can partake of the cultural power residing in scientific knowledge. In so doing, technical writing participates in a writing that conquers naive knowledge by educating it into the technologies of scientific disciplines. Thus, technical writing participates in a system of knowledge and power within our culture (conceived as institutional relationships extending beyond the walls of any one organization). Technical writing also enables technology users to

participate in our culture's knowledge/power system—a system that can be illuminated and analyzed using critical approaches to discourse practices. If, for example, technical writers' practices were studied within one organization (as in social constructionist research designs), the resulting analysis would not be able to show how the texts these writers produced were shaped by—and worked to shape—texts in related institutions, such as government regulations, competitors' work, economic policies, laws, and so on. Researchers can analyze these textual workings of our culture's knowledge/power system, however, using a cultural studies design.

Cultural Studies in Technical Writing

If social constructionist research in technical and professional communication cannot illuminate issues of knowledge and power, some recent cultural studies illustrate how these issues can be explored using an extended idea of culture. In a study based on poststructural theory, Richard Freed advocates widening our views of discourse communities beyond “company specific” boundaries to enable analysis of inter-company relationships (213, n. 8), thereby bringing corporate relationships into play and expanding culture beyond one organization. Freed's work also illustrates how using Lyotard's notions of grand narratives and *petit recits* as set out in *The Postmodern Condition* enables researchers to reconceptualize knowledge as historically situated: “Because the shape and tonality of knowledge vary by locale, and because for that locale the tone and temper of its knowledge rings and feels true, truths at one locale may be different from those held self-evident at other sites and from those held at different times at the same locale” (204). Freed's work further illustrates how using Baudrillard's notion of simulacra problematizes roles of texts, authors and subjects within professional communication:

[I]n a world of copies, of simulations without origins, originality means assembling copies in a new way, like recombinant DNA, by taking existing bits and bytes of text and recombining them. Thus not only is the author already written, by the prescribed roles that the organization or group authorizes him to play, but his materials for discourse are already inscribed, in the intertextual system that allows him to speak. This so-called death of the author, however, doesn't mean his demise, only that the scribe is always already circumscribed. (212)

Use of critical theory allows Freed to explore how technical and professional communication work with situated knowledge that is profoundly shaped by contests for legitimation and which, in turn, shapes subsequent discourse and knowledge.

Other researchers have begun to consider how discourse practices participate in institutional relationships. For example, Bruce Herzberg advocates that composition researchers use Foucault's archaeological research approach "to analyze more closely the role of our institutions and disciplines in producing discourse, knowledge, and power" (80). He asks this question of the social constructionist research paradigm: "[W]hen the group agrees on standards for sufficient evidence or adequate organization or coherent argument, what is the source of its authority?" (79). This questioning of the basis of authority opens up discussions of how discourse participates in power/knowledge systems. In other words, what power sanctions the authority of the knowledge that is described in observed discourse practices? Ben and Marthalee Barton also ask this question of power and authority in exploring the practice of cartography. They found that maps as discourse practices were closely linked to institutional systems of power and knowledge: "Ultimately, the map in particular and, by implication, visual representations in general are seen as complicit with social-control mechanisms inextricably linked to power and authority" (53). These two examples of institutional critique illustrate how discourse can be seen as participating in systems of knowledge and power, exploring why some knowledge is articulated and legitimated while other possible knowledge is marginalized or left silent.

In another example of how critical theory can inform writing research, Lester Faigley explores how subject positions we constitute in composition classrooms rely on teachers' roles "as representatives of institutional authority" (130). By applying Foucault's archaeology method, Faigley finds a technology of confession in writing classrooms where personal narratives are seen as productive of "truths." This technology reproduces existing relations of knowledge and power between teachers and students, in which teachers dominate and students are dominated: "Such an assignment of authority through a teacher's claim to recognize truth is characteristic of Foucault's description of the modern exercise of power. Foucault writes that power is most effective when it is least visible" (131). Using Foucault's archaeology enabled Faigley to discuss issues of institutional power and individual subject positions in the composition classroom because Foucault's work, and critical theory in general, provides a theoretical basis for recognizing institutional relationships and a vocabulary for discussing power/knowledge systems.

Recent cultural studies of technical writing and composition point to the fruitfulness of an approach based on Foucault's archaeological research methods and augmented by closely related lines of critical theory. Applied to technical writing, this approach can illuminate how struggles for knowledge legitimation taking place within technical writing practices are influenced by institutional, political, economic, and/or social relationships, pressures, and tensions within

cultural contexts that transcend any one affiliated group. This type of cultural study can help to answer questions about why technical writing practices work to legitimate some types of knowledge while marginalizing other possible knowledges.

In order to look at struggles for knowledge legitimation that take place within technical writing, researchers can begin by asking Foucault's question, "How is it that one particular statement appeared rather than another?" (*Archaeology* 27). The statements that did appear in technical texts retell stories of the struggles, contradictions, and tensions within historic relations of knowledge and power. These statements also hold the silence of other statements that were possible but did not appear in technical texts at the particular time and place under study. By looking at statements that did appear and positing possible statements that did not appear, the genealogical researcher can construct what Foucault called a "systematic history of discourses" (*Birth* 14). Such a systematic history (or genealogy) of discourse asks questions about how one possible discourse was produced and legitimated as knowledge through technical writing, while other possible discourses were not produced and legitimated. This question of how one group's discourse became knowledge within a situated culture while another group's discourse was not seen as knowledge strikes at the heart of current discussions within technical and professional communication: multiculturalism (Bell, Dillon, and Becker; Kossek and Zonia; Nicholson et al.; Sauer; Zak), postmodernism (Barton and Barton; Britt, Longo, and Woolever; Freed), gender issues (Boiarsky et al.; Dragga; Griffeth et al.; Lay, "Feminist," "Gender," and "Value"), conflict (Burnett; Herndl; McCarthy and Gerring), and ethics (Doheny-Farina; Katz; Porter). Therefore, issues such as multiculturalism, postmodernism, gender, conflict, and ethics within technical and professional communication can be analyzed using a cultural studies research design.

What About Objectivity? What About Validity?

Adopting a cultural studies approach leaves a researcher open to criticism that the results of this type of speculative research are not objective and, therefore, not valid. In other words, how are the results of cultural studies research to be understood in comparison to the results of scientifically modeled research? Choosing a coherent theoretical framework for a cultural study is crucial to understanding its results.

Cultural studies has been described by Cary Nelson, Paula Treichler, and Lawrence Grossberg as an "aggressively anti-disciplinary" activity drawing on many bodies of theory and having no distinct methodology (2). If this is the case, researchers who set out to do

cultural studies must carefully construct theoretical foundations to underpin their work, since there is no ready-made disciplinary or theoretical foundation for this type of study. Researchers can choose from many bodies of theory and/or methodology for constructing these foundations, such as anthropology, sociology, feminism, (neo-)Marxism, or poststructuralism, to list a few possibilities. In the process of choosing a body of theory and/or methodology to work with and setting out a theoretical foundation for the work, researchers will necessarily reflect on the nature of the study and their relationship to the object of inquiry. In this process, researchers will develop explicit rationales for choosing a body of theory to form a foundation for the cultural study. By considering technical writing practice and the research question within a context, cultural researchers cannot hope to adopt research stances outside the context of the object of inquiry. Technical writing must be studied within its context of institutional relationships, which includes researchers as only one of many potentially "confounding" factors.

This relationship of researchers to the object of cultural study bears closer scrutiny, since this relationship precludes claims of "objectivity" which are most often goals of modern academic study. In using this concept of "objectivity," I intend the term in a realist or positivist sense, in which it means something like this: observing the phenomenon without interference from (confounding) human factors. Other definitions of "objectivity" are possible. For one instance, pragmatists might define "objectivity" as sufficient intersubjective agreement for the phenomena to be viewed as "real" rather than as subjective. A postmodern view might see pragmatic "objectivity" as opposed to the realist "objectivity" and, therefore, participating in the modernist search for a metaphysical idea of "objectivity" through a progressive science or philosophy. In responding to these modernist notions, a postmodern view might define "objectivity" as impossible or irrelevant. (See Richard Rorty's "Science as Solidarity" and Hugh Tomlinson's "After Truth" for a discussion of pragmatic and postmodern views of objectivity.)

If objectivity is not possible, how is a cultural study to be considered "valid" by other academic researchers, especially those who conduct research in the quantitative model? How are readers to understand and evaluate the results of a cultural study?

The decentered, unstable author position resulting from self-reflexive poststructural research necessarily impacts traditional ideas of validity. Instead of asking, like Don Stacks and John Hocking, "whether a measurement technique . . . provides measures of what its user thinks it is measuring" (127), a poststructural view of validity would ask (following D. Gordon or Patti Lather) whether the research is an "incitement to discourse" (Lenzo 19), that is, whether the research prompts further discourse as potential objects of future

research. This further discourse would indicate that researchers and other interested parties considered the findings relevant for continued conversations. In light of this self-reflexive and poststructural reconstruction of the idea of validity, a cultural study can be understood as situated—a questioning of representation and reality as they have been constructed at a certain place and time. This idea of validity also suggests that a cultural study will be partial and influenced by the researcher who undertakes the research. In other words, a poststructural cultural study can be “valid” according to a revised and enlarged notion of validity, but it will not be an “objective” study.

The assertion that a cultural study can achieve validity without objectivity flies in the face of centuries of scientific research—a radical position not to be adopted without understanding the scientific tradition against which it is positioned. The scope of this paper certainly does not permit a discussion of the development of the Western scientific tradition to trace thoroughly the tension between science and speculation into which cultural studies must be placed. It is important to trace this tension, however incompletely, because it is an example of how our culture legitimates scientifically derived knowledge while marginalizing knowledge derived through speculation (such as practiced in cultural studies).

To understand cultural studies results in relation to results of more scientifically modeled research, I will briefly trace a contest for knowledge legitimation between experiment and speculation to the works of Francis Bacon, who advocated a natural philosophy based on experience and argued against a philosophy based on logical speculation in the scholastic tradition. Bacon is generally credited in histories of Western thought with articulating an argument for scientific knowledge based on methodical observation of natural phenomena, with the “systematic reform of natural philosophy” (Creighton viii), or with seeking “to reform the whole of human learning” (Zapfen 61). Bacon can also be seen (from a poststructural point of view) as caught up in a contest for knowledge legitimation being waged between the dominant Scholastics and the upstart Reformationists. Francis Bacon in one sense sought to free “truth” from the hold of the Christian Church and a Scholastic tradition based on Aristotelian logic. Bacon’s project for a public science promised to improve the human condition through a methodical, rational study of Nature. But in focusing on a public science and an improved human condition, Bacon necessarily addressed social and political concerns. Bacon’s social project is concerned with reforming and securing the governance of the state as much as with improving the condition of its inhabitants. It is an art of managing people that Bacon sets out as much as a project for dominating Nature.

From our vantage point, we can see that Bacon’s project for a modern, method-driven public science has been successful. This project, begun three centuries ago, overthrew the once-dominant

scholastic tradition. In accomplishing this overthrow, the authority of modern science replaced the once-dominant logical speculation with the now-dominant scientific method. Scientists have accomplished this replacement by their appeal to utility and progress for the improvement of the human condition and they have, indeed, produced improvements in living conditions. Because of these improvements, people who deal in speculative or other non-dominant ways of knowing agree to participate in the hegemony of the sciences.

Those of us living in the United States at the late twentieth century take for granted the dominant place of science in our culture. We often assume that science gives us objective truths. We do not readily see scientific dominance as an outcome of contests for knowledge legitimation that came to a head in our culture some three hundred years ago—and are still waged today.

When scientific discourse is seen as producing objective truths, it is not seen as participating in culturally contextualized contests for dominance and power. In order to see how scientific and technological discourses participate in cultural contests, Stanley Aronowitz argued that these discourse practices must be placed in cultural contexts that extend beyond laboratories or one company's research and development department: "The point is . . . to situate science as a discourse within a larger system of social relations in which economic and political influences do not necessarily appear directly in the laboratory" (19). Since technical writing practices are so closely related to science, the same could be said about them: the point is to situate technical writing as a discourse within larger systems of cultural relations which do not necessarily appear directly wherever the practice takes place. These larger cultural relations may not appear directly because we assume many of them to be natural states of affairs, invisible relations enmeshed in intricate webs of institutional influences that appear inevitable.

One point of studying technical writing as a cultural practice is to make visible what seems invisible in technical writing, to view what seems inevitable as a product of culture. But since this approach to studying technical writing could potentially include everything in its situated cultural context, the researcher needs to limit the study by constructing a coherent theoretical framework for focusing the study and describing the object of inquiry in its institutional relationships.

Delimiting the Object of Cultural Study

If a cultural study can potentially include everything within a culture, how can a researcher construct a framework from which to study technical writing? If a cultural study cannot claim objectivity and relies on a test of validity that calls for the chosen methods to "incite discourse," what methodology can a researcher use to guide the construction of this framework?

Unlike quantitative research projects which can rely on standard methods to guide the research design, cultural study remakes guidelines anew each time. Instead of relying extensively on standards, a cultural researcher relies on analysis of the situatedness of technical writing practices within a cultural context (which includes the researcher) to guide the research design. In the absence of scientific standards and methods, a cultural researcher relies on self-reflexive and non-scientific approaches to understanding the object of inquiry within a cultural context. Underlying this self-reflexive approach to poststructural cultural study is a philosophy that people know the world through language and that we construct realities through language. Because of this primacy of language in poststructural theory, cultural studies focus on discourse practices (such as technical writing) as objects of study.

In designing a cultural study of technical writing, a researcher can reflect on the following five (of many possible) themes: the object of the study is discourse, the object is studied in its cultural context, the object is studied as historically situated, the object is ordered by the researcher for the purposes of the study, and, therefore, the most important relationship in the study is between the object and the researcher. By reflecting on concepts such as these to design a cultural study, the researcher will be building a coherent framework of poststructural theory. The following brief discussions of these five themes provides a place to start thinking about their impact on technical writing research design.

The Object as Discourse

The concept of discourse as an object of inquiry is fundamental to work in cultural studies, wherein knowledge is viewed as constructed and legitimated through language practices. For this reason, technical writing is an especially apt object for cultural study. In *The Archaeology of Knowledge*, Foucault focuses on the concept of discourse as an object of inquiry in setting out his archaeology methodology, describing it as “a systematic description of discourse-objects” (140). The purpose of an archaeological study is to describe how rules of ordering statements both construct knowledge that appears in these statements and subjugate other possible knowledges which remain absent from the statements. François Lyotard in *The Differend* and de Certeau in *Heterologies* further explore how non-legitimated practices struggle with legitimated language practices to be heard within hegemonic social and political contexts. This consideration of the discourse object within complex cultural contexts is the basis of cultural studies. Since technical writing is clearly discursive, it can be readily constituted as an object of cultural study according to this theoretical concept.

Object within Cultural Context

Because technical writing is a practice that takes place within the context of dominant and subordinated knowledges (scientific and humanistic traditions, for example), constituting it as an object of cultural study can illuminate its role in complex social, economic, political, and institutional relations, which cannot be seen as well using quantitative or social constructionist methods. The study of cultural objects within their contexts is a practice of cultural studies which can be traced through early works in this field. Work done in the Birmingham Centre and reported by Stuart Hall et al. in *Culture, Media, Language* can be seen to reflect both the influence of Antonio Gramsci's writings in the *Prison Notebooks* and his notions of hegemony, combined with the Centre's earlier anthropological influences. From this early work, cultural studies has adopted anthropological notions that objects of inquiry can include everyday practices that were not seen as significant in the knowledge-making of some other disciplines. In *The Practice of Everyday Life*, for example, de Certeau clearly illustrated the importance of studying the contributions of mundane practice, explaining how people can accomplish non-legitimated actions within a dominant culture. Technical writing can be seen as one of these mundane practices, whose role within a cultural context can be more completely described using a cultural studies research methodology in tandem with other more scientifically modeled methods.

Object within Historical Context

Because technical writing practices can be considered as being shaped by—and shaping—their cultural contexts, a cultural study of these practices will select significant moments in history to describe the relationships between technical writing and the surrounding culture. In effect, this type of study will often take the form of a descriptive history about the object of inquiry at a specific time and place. In *The Birth of the Clinic*, Foucault calls for studying functional segments of discourse to create a systematic history which does not assume any remainder of meaning in the discourse. In other words, his purpose is to describe discourse-objects in their historical cultural contexts, without attempting interpretations of the objects. In *The Writing of History*, de Certeau also argues that discourse should be studied in its historical aspects and as resulting from practices. He further saw the impulse that prompted the historical study as originating in current events. We may study cultural objects in their local, historical contexts, for example, but the impetus for the study exists in our present relations with the cultural objects. This concept tells us that technical writing can be constituted as an object of cultural study by placing it in an historical context and describing it within that

context. The purpose of this type of study is not to interpret explicitly the role of the object within the context, but more to set out a history of that relationship of object and context, acknowledging that any discourse involves interpretation. The history a cultural study seeks to represent includes, as much as possible, subjugated knowledges as well as dominant ones.

Object as Ordered

In all the cultural studies mentioned so far, Benjamin would have said that the one who undertook the study imposed an ordering on the object and its contexts, thereby including some ways of understanding the object and excluding others. He elaborates on this theme in "Theses on the Philosophy of History" when he argues that cultural objects are the spoils of the victor, but still hold within their silences the barbarism of the victory. This theme is reflected in many cultural studies' concern for difference, knowledge legitimation, and otherness. In *Power/Knowledge*, Foucault clearly acknowledges that some ways of knowing are subjugated, but are still present in language even in their ostensible absence. He called for us to create difference in discourse as an object of study, to uncover the subjugated knowledge, and then use the subjugated knowledge tactically in practice. In exploring the tactical nature of subjugated knowledge, Foucault describes power/knowledge systems in military terms. In *The Differend*, Lyotard looks at this same type of inclusion/exclusion practice in terms of a legal system, describing a process whereby the linking of phrases, while necessary, is indeterminate as to what types of links can be made. In other words, phrases *must* be linked, but how they are linked is arbitrary and many possibilities for linking exist. In choosing one way of linking, we necessarily exclude other ways, thereby silencing the knowledge that could be made through those other linkings. A cultural study of technical writing would explore those silences, absences, and exclusions still held within the dominant knowledge and discourse of that field's practices.

Relationship of Object with the One Who Studies

The person who researches technical writing practices necessarily orders them for the purposes of the study. Through this process, the researcher includes some possible ways of ordering and making knowledge and excludes others. In the research process, therefore, the most important encounter takes place between the object of study and the person who studies it. Technical writing is constituted in its encounter with the researcher, and it is limited by the researcher. Foucault in *Power/Knowledge* called for constituting the object of study as difference and discontinuity. For de Certeau in *Heterologies*, the aberrant

can suggest an object for study. But whatever the researcher constitutes as an object of inquiry, remember Benjamin's observation that objects of cultural study, such as technical writing, are caught in a dialectic tension of chaos and order, present and past, rational and nonrational. Cultural researchers seek to describe technical writing while recognizing their participation in these dialectical tensions that affect professional practices and the study.

Conclusion

A cultural study describes an anti-disciplinary, situated, and personal encounter between a researcher and technical writing practices. When this study is firmly grounded in coherent theoretical rationales and employs a rigorous methodology, the speculative knowledge gained through the research can be considered valid in an expanded sense of that notion. By adding cultural studies to our more scientifically modeled research in technical writing, we can add discussions of power, politics, ethics, and cultural tensions to our understandings of what it is we do when we communicate. With an expanded idea of culture, we can expand our understanding of technical writing practice.

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